

SwarupWang_ForecastCompetitionReport

GitHub

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Data Wrangling

Load load, temp, and humidity files

```
# Load the demand excel file
load_df <- read_excel("../Data/load.xlsx")

daily_load <- load_df %>%
  group_by(meter_id, date) %>%
  summarize(
    daily_avg = mean(c_across(1:24), na.rm = TRUE),
    .groups   = "drop"
  )

# Load the temperature file
temp_df <- read_excel("../Data/temperature.xlsx")

daily_temp <- temp_df[1:3] %>%
  group_by(date) %>%
  summarize(
    daily_avg = mean(t_ws1, na.rm = TRUE),
    .groups   = "drop"
  )

# Load the humidity file
humidity_df <- read_excel("../Data/relative_humidity.xlsx")

daily_humidity <- humidity_df[1:3] %>%
  group_by(date) %>%
  summarize(
    daily_avg = mean(rh_ws1, na.rm = TRUE),
    .groups   = "drop"
  )
```

load file for humidity and temperature forecasts

The forecasting code can be found in the Code folder

```
daily_temp_for <- read_csv("../Data/Temp_Forecast.csv")
temp_for <- msts(daily_temp_for$x, seasonal.periods =c(7,365.25), start=c(2011,1,1))
```

```
daily_humidity_for <- read.csv("../Data/Humidity_Forecast.csv")
humidity_for <- msts(daily_humidity_for$x, seasonal.periods =c(7,365.25), start=c(2011,1,1))
```

Create Time Series

```
# Create time series for load
load_ts <- msts(daily_load$daily_avg,
               seasonal.periods =c(7,365.25),
               start=c(2005,1,1))

# Create time series for temperature
temp_ts <- msts(daily_temp$daily_avg,
               seasonal.periods =c(7,365.25),
               start=c(2005,1,1))

# Create time series for humidity
humidity_ts <- msts(daily_humidity$daily_avg,
                  seasonal.periods =c(7,365.25),
                  start=c(2005,1,1))

#create a subset for training purpose
n_for = 365
ts_load_train <- subset(load_ts,
                       end = length(load_ts)-n_for)

#create a subset for testing purpose
ts_load_test <- subset(load_ts,
                      start = length(load_ts)-n_for)

# Create date sequence
dates <- seq.Date(from = as.Date("2011-01-01"), to = as.Date("2011-02-28"), by = "day")
```

Best Five Models

Model 1: Forecasting using NN - $p=2$, $P=1$, $K=c(2,4)$

```
K = c(2,4)
for_period = 59

NN_forecast <- function(ts, p, P, K){
  model_fit <- nnetar(ts,
                    p=p,
                    P=P,
                    xreg=fourier(ts, K=K))

  model_forecast <- forecast(model_fit, h=59,xreg=fourier(ts,
                                                         K=K,h=for_period))
}

forecast_1 <- NN_forecast(load_ts, 2, 1, K)

#Plot model + observed data
autoplot(load_ts, color="gray70") +
```

```

autolayer(forecast_1, series="Neural Network",PI=FALSE)+
ylab("Load") +
ggtitle("Daily Load Forecast - Model 1")+
theme_minimal()+
theme(legend.position = "bottom")

```

Daily Load Forecast – Model 1



```

NN_result_1 <- as.data.frame(forecast_1$mean)

# Combine into a dataframe
print_result <- function(result){
  df <- data.frame(Date = dates, Value = result) # Combine into a dataframe
  df_summary <- bind_rows(
    head(df),
    tail(df)
  ) # Get head and tail
  # Print a table
  kable(df_summary, caption = "Model result for the first and last five days") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"))
}

print_result(NN_result_1)

```

Table 1: Model result for the first and last five days

Date	x
2011-01-01	5116.664
2011-01-02	5187.086

2011-01-03	5012.691
2011-01-04	5137.024
2011-01-05	5268.203
2011-01-06	5366.463
2011-02-23	3678.444
2011-02-24	3808.278
2011-02-25	3937.789
2011-02-26	4116.071
2011-02-27	4337.583
2011-02-28	4277.809

Model 2: Forecasting using NN - $p=1$, $P=0$, $K=c(2,4)$

```
K = c(2,4)
for_period = 59

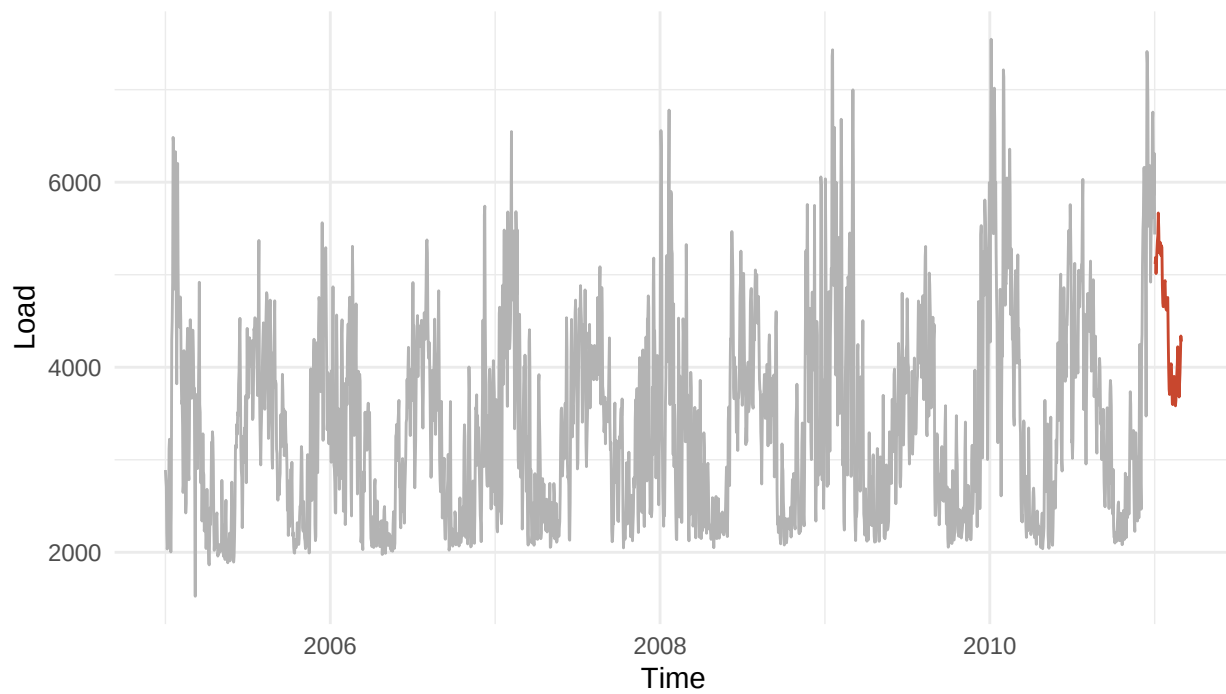
NN_forecast <- function(ts, p, P, K){
  model_fit <- nnetar(ts,
    p=p,
    P=P,
    xreg=fourier(ts, K=K))

  model_forecast <- forecast(model_fit, h=59,xreg=fourier(ts,
    K=K,h=for_period))
}

forecast_2 <- NN_forecast(load_ts, 1, 0, K)

#Plot model + observed data
autoplot(load_ts, color="gray70") +
  autolayer(forecast_1, series="Neural Network",PI=FALSE)+
  ylab("Load") +
  ggtitle("Daily Load Forecast - Model 1")+
  theme_minimal()+
  theme(legend.position = "bottom")
```

Daily Load Forecast – Model 1



series  Neural Network

```

NN_result_2 <- as.data.frame(forecast_1$mean)

# Combine into a dataframe
print_result <- function(result){
  df <- data.frame(Date = dates, Value = result) # Combine into a dataframe
  df_summary <- bind_rows(
    head(df),
    tail(df)
  ) # Get head and tail
  # Print a table
  kable(df_summary, caption = "Model result for the first and last five days") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"))
}

print_result(NN_result_2)

```

Table 2: Model result for the first and last five days

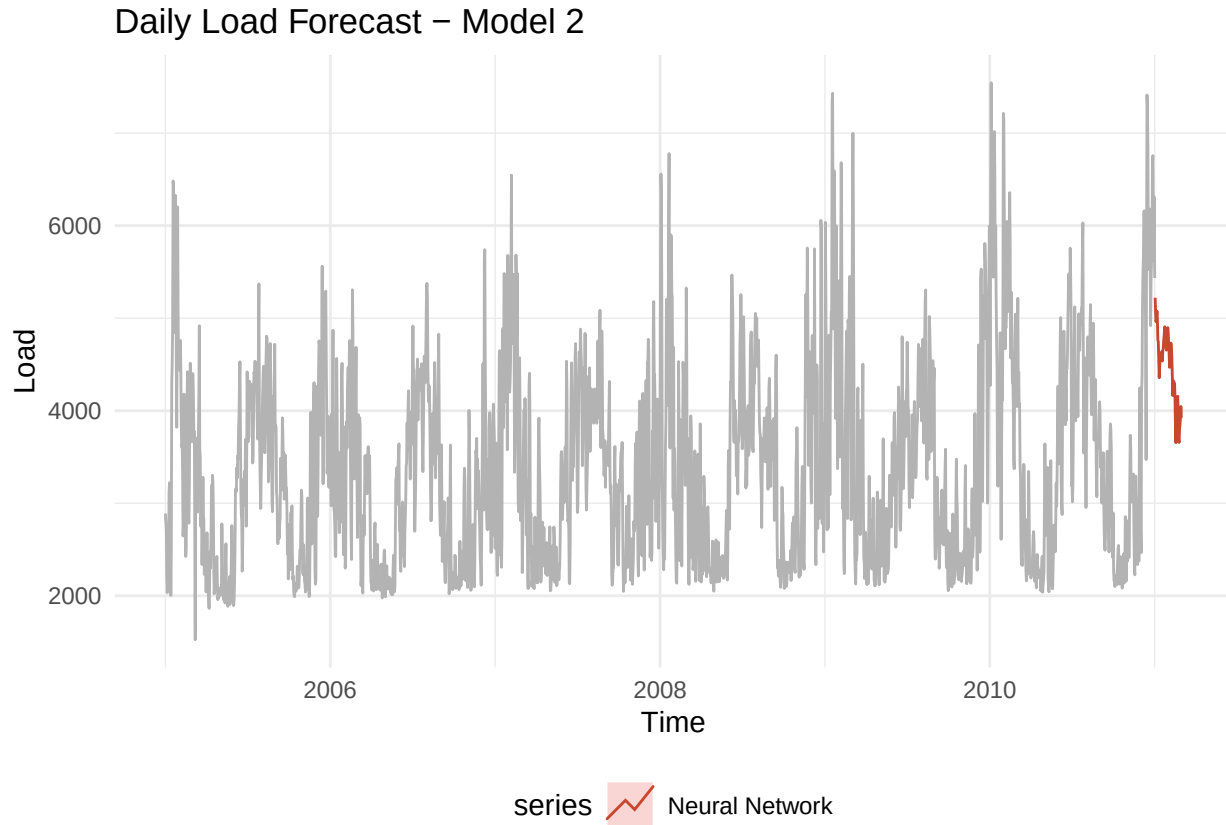
Date	x
2011-01-01	5116.664
2011-01-02	5187.086
2011-01-03	5012.691
2011-01-04	5137.024
2011-01-05	5268.203
2011-01-06	5366.463
2011-02-23	3678.444
2011-02-24	3808.278

2011-02-25	3937.789
2011-02-26	4116.071
2011-02-27	4337.583
2011-02-28	4277.809

Model 3: Forecasting using NN - $p=1$, $P=1$, $K=c(2,4)$

```
forecast_3 <- NN_forecast(load_ts, 1, 1, K)

#Plot model + observed data
autoplot(load_ts, color="gray70") +
  autolayer(forecast_2, series="Neural Network",PI=FALSE)+
  ylab("Load") +
  ggtitle("Daily Load Forecast - Model 2")+
  theme_minimal()+
  theme(legend.position = "bottom")
```



```
NN_result_3 <- as.data.frame(forecast_2$mean)

print_result(NN_result_3)
```

Table 3: Model result for the first and last five days

Date	x
2011-01-01	5221.452
2011-01-02	5071.635

2011-01-03	4962.138
2011-01-04	5004.985
2011-01-05	5075.774
2011-01-06	4981.608
2011-02-23	3657.309
2011-02-24	3806.273
2011-02-25	3854.600
2011-02-26	3920.159
2011-02-27	4046.126
2011-02-28	3923.991

Model 4: Forecasting using TBATS+NN Combined Model with humidity and temperature variables

```
h_future <- 59
K=c(2,4)
# Generate Fourier terms for training (if needed) and for forecasting 2010
fourier_train <- fourier(load_ts, K = K)
fourier_future <- fourier(load_ts, K = K, h = h_future)

# Create xreg matrices for training and forecasting
# (Assumes you have also created ts_temp_train and ts_humidity_train for training period)
xreg_train <- cbind(fourier_train, temp = temp_ts, hum = humidity_ts)
xreg_future <- cbind(fourier_future, temp = temp_for, hum = humidity_for)

# Fit the NNAR model on training data with external regressors
NN_fit <- nnetar(load_ts, p = 1, P = 1, xreg = xreg_train)

# Forecast for h_future days (2010)
NN_for <- forecast(NN_fit, xreg = xreg_future, h = h_future)

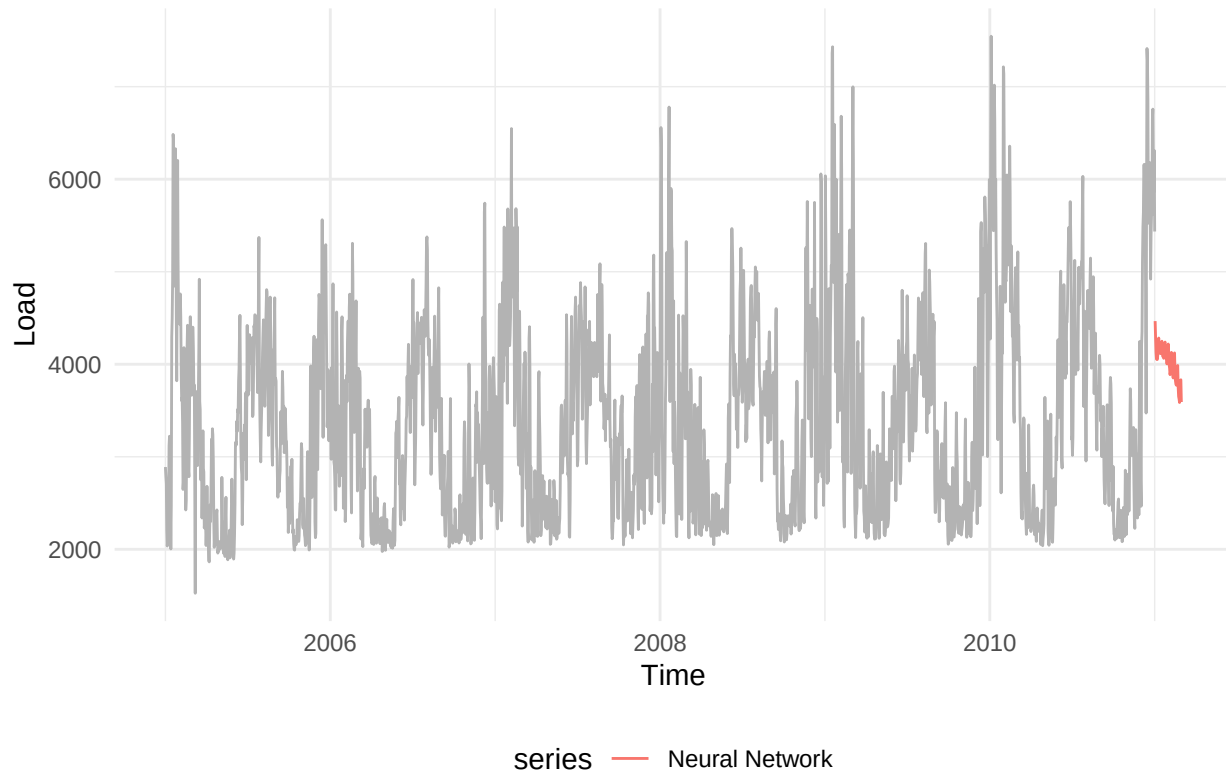
# TBATS can take time to fit
TBATS_fit <- tbats(load_ts)

TBATS_for <- forecast(TBATS_fit, h=59)

hybrid_for <- (TBATS_for$mean + NN_for$mean) / 2

autoplot(load_ts, color="gray70") +
  autolayer(hybrid_for, series="Neural Network",PI=FALSE)+
  ylab("Load") +
  ggtitle("Daily Load Forecast - Model 3")+
  theme_minimal()+
  theme(legend.position = "bottom")
```

Daily Load Forecast – Model 3



```
hybrid_result <- as.data.frame(hybrid_for)
print_result(hybrid_result)
```

Table 4: Model result for the first and last five days

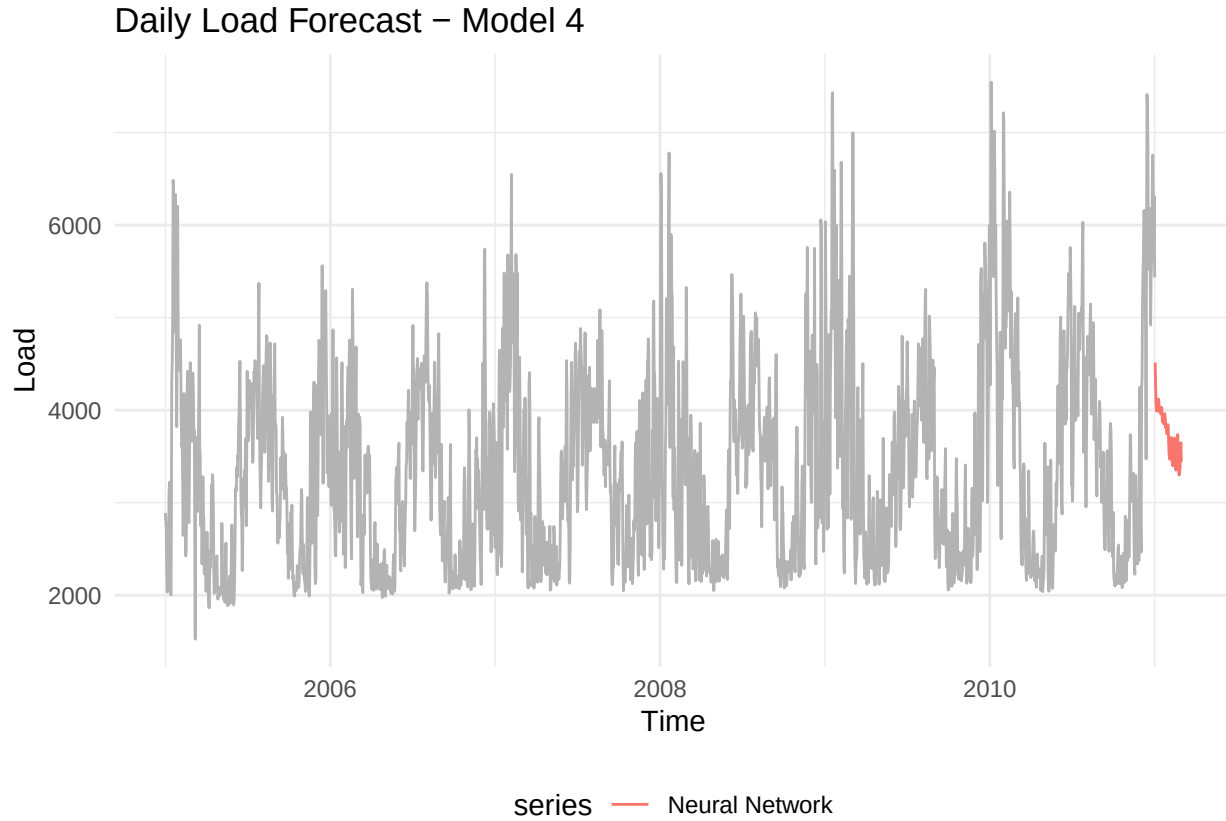
Date	x
2011-01-01	4466.333
2011-01-02	4283.717
2011-01-03	4124.351
2011-01-04	4055.178
2011-01-05	4060.089
2011-01-06	4052.670
2011-02-23	3624.844
2011-02-24	3586.682
2011-02-25	3653.808
2011-02-26	3824.933
2011-02-27	3832.165
2011-02-28	3591.673

Model 5: Forecasting using further combined model based on Model 3

```
combined_for <- (TBATS_for$mean + forecast_3$mean) / 2
forecast_5 <- (combined_for + NN_for$mean) / 2
```



```
autoplot(load_ts, color="gray70") +
  autolayer(forecast_5, series="Neural Network",PI=FALSE)+
  ylab("Load") +
  ggtitle("Daily Load Forecast - Model 4")+
  theme_minimal()+
  theme(legend.position = "bottom")
```



```
model5_result <- as.data.frame(forecast_5)
print_result(model5_result)
```

Table 5: Model result for the first and last five days

Date	x
2011-01-01	4513.754
2011-01-02	4266.612
2011-01-03	4061.931
2011-01-04	3992.596
2011-01-05	3990.014
2011-01-06	3986.644
2011-02-23	3300.036
2011-02-24	3315.138
2011-02-25	3376.158
2011-02-26	3553.020
2011-02-27	3647.976
2011-02-28	3443.750